Laptop not charging or battery draining fast

Article KB0010036 | Owner: Desktop Support | Subcategory: Battery / charging | Applies to: Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840, MacBook Pro/Air, Windows 11 23H2/24H2, macOS 14/15

Summary

This article helps users diagnose why their Acme-issued laptop is not charging or the battery is draining unusually fast. It outlines quick self-checks, a step-by-step self-help guide, and the process Desktop Support follows to isolate hardware faults, replace chargers or batteries, and safely handle swollen cells.

Symptoms

- Battery icon shows 0% while the AC adapter is plugged in
- Charging LED on the laptop does not illuminate when connected
- Battery percentage drops more than 10% per minute while the system is idle
- OS reports "Plugged in, not charging"
- Visible swelling, bulging or deformation of the battery pack
- Charger feels excessively hot, emits a burning smell, or makes unusual noises

Common causes

- Faulty AC adapter, power cable, or connector
- Damaged or dirty charging port on the laptop
- Battery health degraded below ~60% design capacity
- Swollen battery due to age, over-charge, or manufacturing defect
- Incorrect power-management settings or outdated BIOS/firmware
- Intune compliance policies that temporarily block charging
- Corrupted battery driver or EC (embedded controller) firmware

Preliminary Checks

Before proceeding with troubleshooting, perform a few quick visual and functional checks. These steps help rule out obvious issues such as a loose cable or a non-functional power outlet. If any of the checks reveal a clear problem (e.g., frayed cable, no LED light), you can move directly to the self-help procedure or create a ticket for Desktop Support.

Self-Help Procedure

Follow these steps to isolate the most common charging problems. Perform each step in order and test after each action. If the laptop begins to charge normally, stop and monitor the battery for a few hours to confirm the issue is resolved.

1. 1. Disconnect the charger from both the laptop and the wall outlet. Inspect the AC adapter brick, cable, and connector for visible damage, kinks, or burnt smells. If damage is found, note it and proceed to step 4.
2. 2. Plug the charger into a different known-good wall outlet (preferably on a different circuit). Observe whether the charging LED lights up or the battery icon changes.
3. 3. If the charger uses a detachable cable, try a spare AC-adapter cable if one is available. Otherwise, continue to the next step.
4. 4. With the charger connected, press and hold the laptop's power button for 15 seconds to perform an embedded controller (EC) reset. Release the button, wait 10 seconds, then attempt to power on the laptop.
5. 5. Boot into the BIOS/UEFI (usually by pressing F2, Del, or Esc during startup). Check the battery information page for reported design capacity vs. full charge capacity. Note any large discrepancy.
6. 6. For Windows laptops, open Device Manager -> Batteries, right-click "Microsoft ACPI-Compliant Control Method Battery" and select "Uninstall device". Confirm uninstall, then shut down, disconnect the charger, wait 30 seconds, reconnect the charger and power on - Windows will reinstall the driver.
7. 7. For macOS, reset the System Management Controller (SMC): shut down, press Shift-Control-Option on the left side and the power button simultaneously for 10 seconds, release, then power on.
8. 8. After completing the above, observe whether the laptop charges normally and whether the battery percentage increases over the next 30-60 minutes.

Service Desk Procedure

When a user creates a ticket or the self-help steps do not restore charging, Desktop Support agents follow this structured workflow. The goal is to identify whether the fault lies with the charger, the charging port, or the battery itself, and to take the appropriate remedial action while observing safety precautions for swollen cells.

1. 1. Verify ticket details: capture the exact symptom text, laptop model, serial number, and any error messages (e.g., "Plugged in, not charging"). Ask the user to confirm whether the charger LED illuminates when plugged in.
2. 2. Perform a physical inspection of the charger and cable: look for frayed insulation, bent pins, or signs of overheating. If defective, replace the charger with a known-good spare from inventory and test.
3. 3. If charging still fails, inspect the laptop's charging port for debris, corrosion, or mechanical damage. Clean gently with compressed air if needed; if the port is damaged, escalate to the Desktop Support lead for possible motherboard RMA.
4. 4. Using the BIOS/UEFI or the manufacturer's diagnostics tool (Dell BIOS, Lenovo Vantage, HP Support Assistant), check the reported battery health (design capacity vs. full charge capacity). Record the percentage.
5. 5. If battery health is below 60% or the battery reports "Replace soon", order a replacement battery through the standard hardware provisioning process. Follow the laptop-specific battery replacement guide (power down, disconnect charger, remove bottom cover, disconnect battery connector, install new battery, reassemble).
6. 6. If the battery appears swollen, bulging, or leaking, treat it as a safety hazard: immediately shut down the laptop, disconnect the charger, place the device on a non-conductive surface in a well-ventilated area, and label it "Swollen Battery - Do Not Use". Notify the Desktop Support lead and the Security Operations team for safe disposal and arrange a replacement unit.
7. 7. After replacing the charger or battery, plug the laptop in and verify that the charging LED illuminates and the battery percentage rises. Run a brief stress test (e.g., play a video for 15 minutes) to confirm stable charging.
8. 8. Document the parts used, the serial numbers of replaced items, and close the ticket with a resolution note indicating whether the issue was charger-related, battery-related, or port-related.

Verification and Closure

After any hardware replacement, it is important to confirm that the laptop returns to normal operation and that the user is satisfied. This section outlines the final checks agents should perform before closing the ticket.

1. 1. Ask the user to confirm that the battery charges to 100% and holds charge for at least 2-3 hours of typical use (web browsing, Office apps).
2. 2. Verify that the battery health reported in BIOS/UEFI or the vendor utility shows a capacity of 80% or higher for a new battery.
3. 3. Ensure that the laptop powers on and off correctly with both AC power and battery only.
4. 4. For Windows devices, run the built-in powercfg /batteryreport command and review the recent capacity values; for macOS, check System Information -> Power.
5. 5. Close the ticket in the service-desk system, adding the resolution summary and any parts replaced. Tag the ticket with the appropriate category (Hardware -> Battery -> Charging).

When to escalate

If the battery is physically swollen, leaking, or emits smoke, treat it as a safety incident: immediately quarantine the device, notify the Desktop Support lead and Security Operations, and follow the hazardous-material handling procedure. If after two charger replacements and one battery replacement the laptop still fails to charge, escalate to the Desktop Support senior engineer for possible motherboard or charging-circuit fault, providing the laptop's serial number, charger part numbers, and diagnostic logs.

Keywords: laptop not charging, battery draining fast, charger not working, battery swollen, plugged in not charging, battery health low, power adapter faulty, charging port issue, EC reset, SMC reset, battery replacement, swollen battery safety